NagarMitram: A Smart City Solution for Civic Issue Management

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Abstract

Rapid urbanization has increased the number of civic issues such as potholes, garbage accumulation, water leakage, and damaged public infrastructure, creating significant challenges for municipal authorities. Traditional complaint management systems are often slow, inefficient, and lack transparency, resulting in delayed issue resolution and reduced citizen satisfaction. This research presents **NagarMitram**, an AI-powered civic issue reporting and resolution system designed to enhance communication between citizens and municipal departments through a digital platform.

The proposed system enables users to report civic issues using geotagged photographs, GPS-based location tracking, and detailed descriptions through a mobile application. The platform incorporates artificial intelligence techniques for automatic issue classification, severity assessment, and intelligent routing to the appropriate department. A web-based administrative dashboard provides real-time monitoring, analytics, and issue management capabilities for municipal authorities.

The implementation utilizes modern technologies including React Native, Next.js, TypeScript, PostgreSQL, and AI integration tools. Experimental evaluation demonstrates improved complaint handling efficiency, faster response times, enhanced transparency, and better citizen engagement compared to traditional methods. The study concludes that AI-driven civic management platforms can significantly improve urban governance by promoting accountability, data-driven decision-making, and active citizen participation. NagarMitram provides a scalable and intelligent framework for developing smarter, more responsive, and citizen-centric cities.

Keywords **-** Civic Issue Management, Artificial Intelligence (AI), Smart Governance, Mobile Application, Civic Complaint Reporting, Issue Classification, GPS-Based Tracking, Urban Governance, Real-Time Monitoring, Citizen Engagement, Predictive Analytics, Smart City, Municipal Services, Data Analytics, Digital Governance.

1. INTRODUCTION

Rapid urbanization and population growth have significantly increased the number of civic issues such as potholes, garbage accumulation, water leakage, damaged streetlights, and sanitation-related problems. Traditional complaint management systems used by municipal authorities are often manual, time-consuming, and lack transparency, resulting in delayed issue resolution and reduced citizen satisfaction.

To address these challenges, NagarMitram is proposed as an AI-powered civic issue reporting and resolution system that enables citizens to report civic problems through a mobile application. The

platform allows users to submit complaints along with photographs, GPS-based location information, and issue descriptions. The reported issues are automatically classified and prioritized using Artificial Intelligence techniques, ensuring efficient routing to the appropriate municipal departments.

The system also provides a web-based administrative dashboard that enables authorities to monitor complaints, track issue status, analyze trends, and manage resources effectively. By integrating real-time data processing, geolocation services, and AI-driven analytics, NagarMitram improves communication between citizens and government agencies while enhancing accountability and transparency.

The project contributes to the vision of smart cities and digital governance by creating a scalable, efficient, and citizen-centric platform for civic issue management.

1.2 KEY OBJECTIVES –

1.2.1 Objective 1

To develop a user-friendly mobile application that enables citizens to report civic issues with photographs, location details, and descriptions in real time.

1.2.2 Objective 2

To implement Artificial Intelligence techniques for automatic issue classification, severity assessment, and intelligent complaint routing.

1.2.3 Objective 3

To provide a centralized administrative dashboard for municipal authorities to monitor, manage, and resolve civic complaints efficiently.

1.2.4 Objective 4

To improve transparency and citizen engagement by offering real-time status tracking, notifications, and feedback mechanisms.

1.2.5 Objective 5

To support smart governance through data analytics, predictive insights, and efficient resource allocation for faster issue resolution.

1.3 PROBLEM STATEMENT –

Urban local bodies and municipal authorities face significant challenges in managing and resolving civic issues such as potholes, garbage accumulation, water leakage, damaged streetlights, drainage problems, and sanitation concerns. Existing complaint management systems are often manual, fragmented, and inefficient, resulting in delayed responses, lack of transparency, poor tracking mechanisms, and reduced citizen trust.

Citizens frequently encounter difficulties in reporting issues, monitoring complaint status, and receiving timely updates regarding resolution. Municipal departments, on the other hand, struggle with prioritizing complaints, allocating resources efficiently, and maintaining effective communication with the public. The absence of a centralized digital platform further leads to data inconsistencies, accountability issues, and ineffective decision-making.

Therefore, there is a need for an intelligent, transparent, and user-friendly civic issue management system that enables citizens to report problems easily while allowing authorities to monitor, analyze, prioritize, and resolve complaints efficiently. NagarMitram addresses these challenges by providing an AI-powered platform that integrates real-time issue reporting, GPS-based location tracking, automated classification, and administrative analytics to improve civic governance and citizen engagement.

2. Literature Review

Author	Contribution	Limitations
Kumar et al.	Civic complaint reporting system	No AI-based classification
Sharma et al.	GIS-based issue tracking	Limited user interaction
Patel et al.	Mobile complaint management app	No predictive analytics
Singh et al.	AI-assisted issue categorization	Limited transparency features
NagarMitram (Proposed)	AI-powered reporting, tracking, and analytics	Future scope for IoT integration

2.1 Research Gap

Existing civic complaint management systems primarily focus on basic issue reporting and tracking. Most solutions lack intelligent issue classification, automated prioritization, real-time analytics, and effective citizen-government interaction. Additionally, many platforms do not provide predictive insights for resource allocation or transparent complaint monitoring.

The proposed **NagarMitram** system addresses these gaps by integrating Artificial Intelligence for issue classification and severity assessment, GPS-based location tracking, real-time notifications, and an administrative dashboard with analytics. This improves efficiency, transparency, citizen engagement, and decision-making in civic governance.

3. Proposed Methodology

3.1 Requirement Analysis

3.1.1 Functional Requirements

3.1.1.1 Requirement 1: The system shall allow citizens to register, log in, and report civic issues with images, descriptions, and GPS-based location details.

3.1.1.2 Requirement 2: The system shall automatically classify and prioritize reported issues using Artificial Intelligence techniques.

3.1.1.3 Requirement 3: The system shall provide an administrative dashboard for monitoring, assigning, tracking, and resolving civic complaints in real time.

3.1.2 Non Functional Requirements

3.1.2.1 Performance

3.1.2.2 Security

3.1.2.3 Reliability

3.1.2.4 Usability

3.2 Object Oriented Analysis

3.2.1 Major Classes

1. User
Attributes: UserID, Name, Email, Password, Role
Methods: register(), login(), logout()
2. Citizen
Attributes: CitizenID, ContactNumber, Address
Methods: reportIssue(), trackIssue(), submitFeedback()
3. Admin
Attributes: AdminID, Department
Methods: assignIssue(), updateStatus(), generateReport()
4. Issue
Attributes: IssueID, Title, Description, Category, Status, Location, DateReported
Methods: createIssue(), updateIssue(), classifyIssue()
5. Department
Attributes: DepartmentID, DepartmentName, AssignedIssues
Methods: receiveIssue(), resolveIssue()
6. Feedback
Attributes: FeedbackID, Rating, Comments, Date
Methods: submitFeedback(), viewFeedback()
7. Notification
Attributes: NotificationID, Message, DateTime
Methods: sendNotification(), markAsRead()

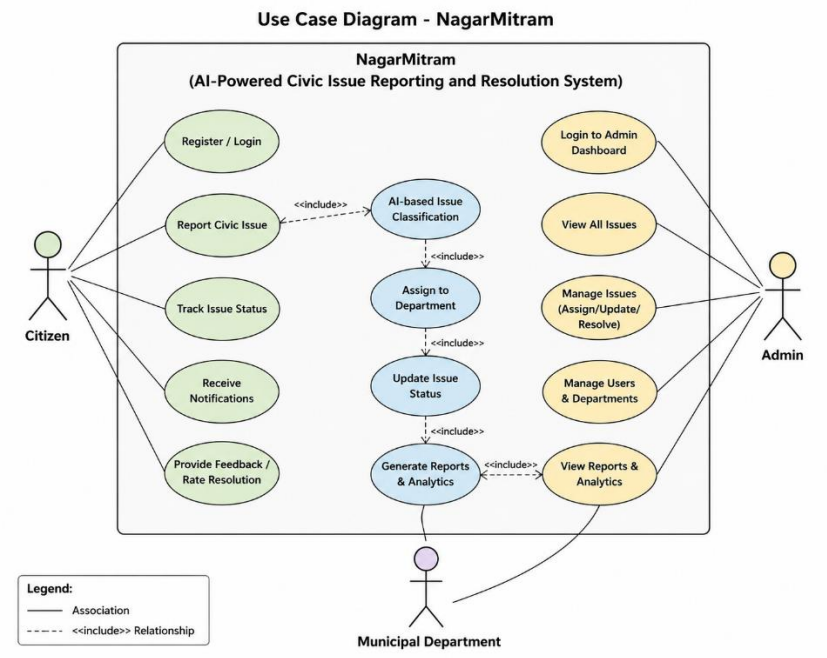
3.2.2 Relationships among classes

3.2.2.1 Association
3.2.2.2 Aggregation
3.2.2.3 Composition
3.2.2.4 Inheritance

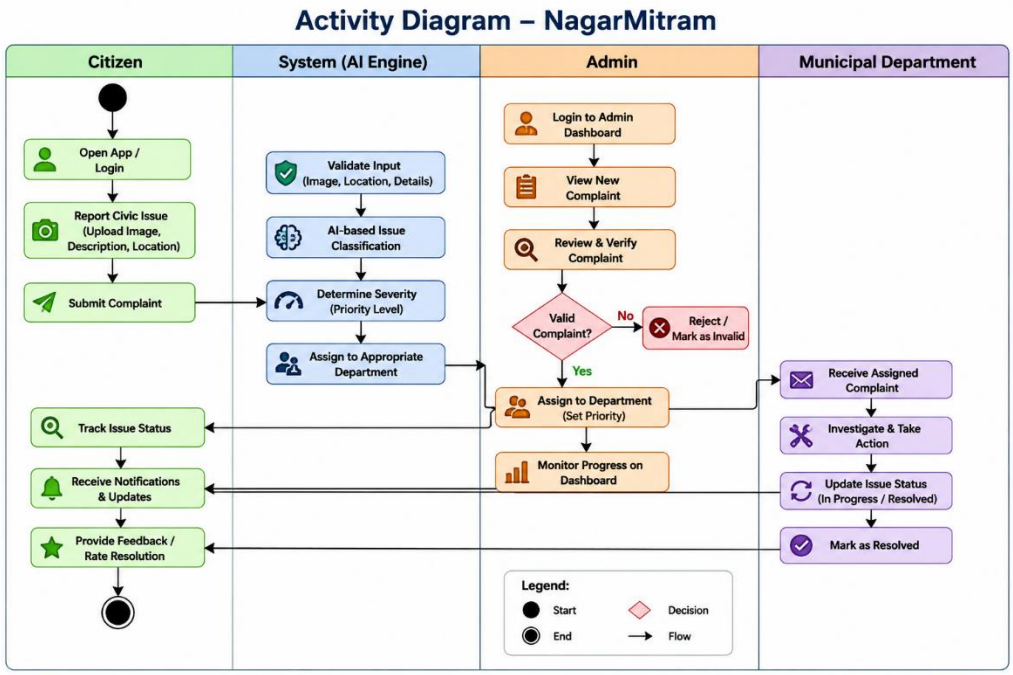
3.3 UML

Develop the following UML diagrams:

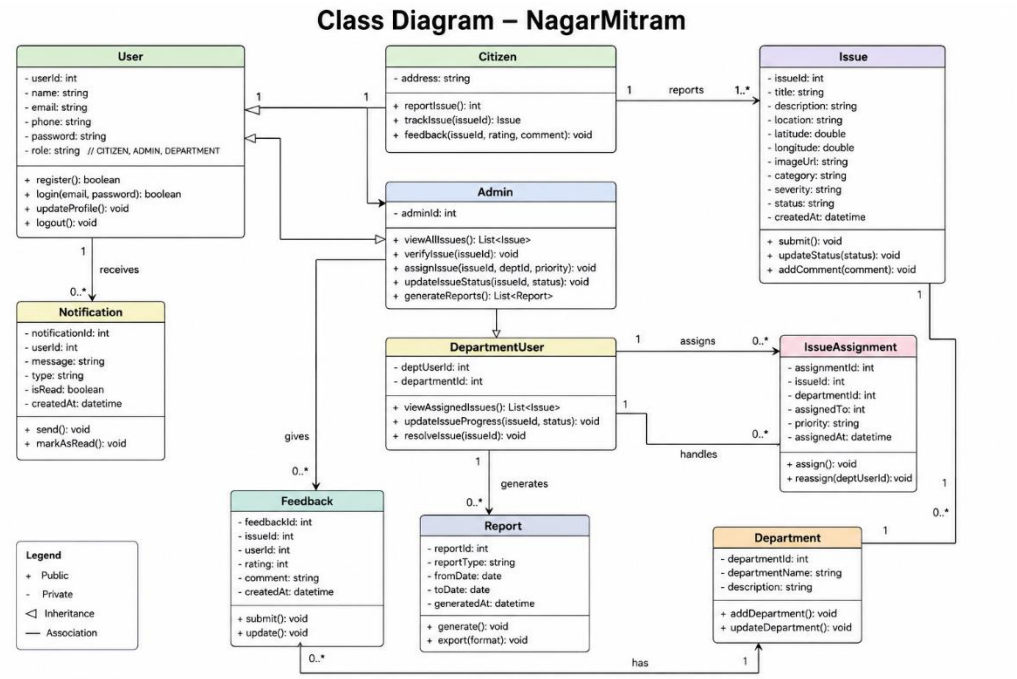
- Use Case Diagram



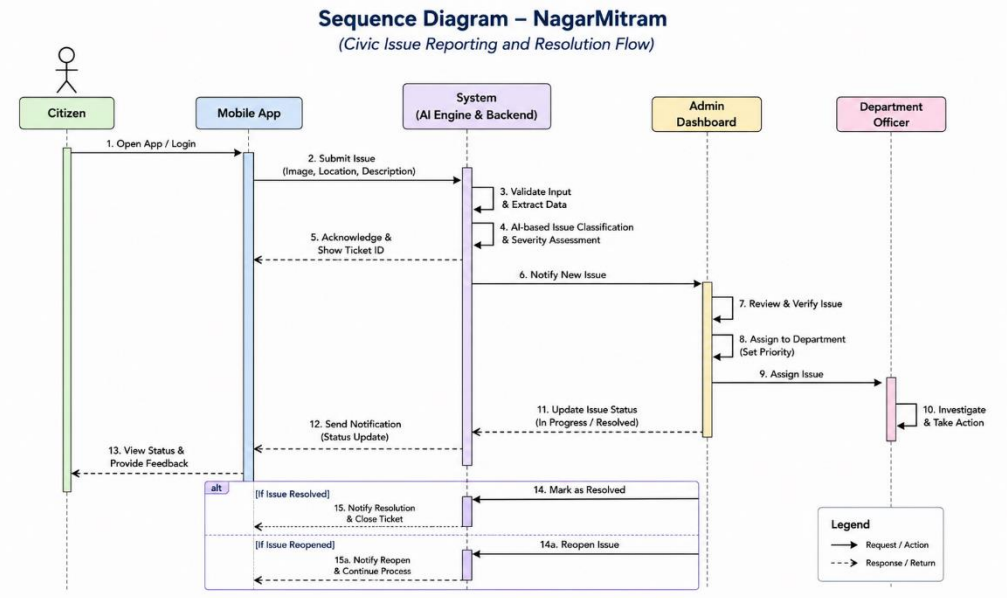
- Activity Diagram



- Class Diagram



- Sequence Diagram



4 Design & Implementation:

4.1 OO Concepts Used

4.1.1 Encapsulation

Encapsulation is implemented by keeping class attributes private and providing controlled access through public methods (getters and setters). For example, user details, issue information, and department data are stored within their respective classes and can only be

accessed or modified through authorized methods. This ensures data security, integrity, and controlled access to sensitive information.

4.1.2 Abstraction

Abstraction is achieved through the use of interfaces and service layers that hide implementation details from users. For example, authentication services, issue management services, and notification services expose only the required functionalities while concealing the underlying database operations and AI processing logic. This simplifies system usage and improves maintainability.

4.1.3 Inheritance

Inheritance is used to establish parent-child relationships among classes. The **User** class acts as a base class containing common attributes such as User ID, Name, Email, and Password. Specialized classes such as **Citizen**, **Admin**, and **DepartmentUser** inherit these common properties and extend them with role-specific functionalities, reducing code duplication and improving reusability.

4.1.4 Polymorphism

Polymorphism is implemented through method overriding and dynamic behavior. Different user roles perform actions using methods with the same name but different implementations. For example, the login() method may behave differently for Citizens, Admins, and Department Users based on their permissions. Similarly, issue handling and notification methods can be customized according to user roles, providing flexibility and extensibility in the system design.

4.2 SOLID Principles Applied

4.2.1 Single Responsibility Principle
4.2.2 Open-Closed Principle
4.2.3 Liskov Substitution Principle
4.2.4 Interface Segregation Principle
4.2.5 Dependency Inversion Principle

4.3 Design Patterns Used

4.3.1 Singleton Pattern

The Singleton Pattern is used to maintain a single instance of critical services such as Database Connection Manager and Authentication Service throughout the application lifecycle. This reduces resource consumption and ensures consistent access to shared resources.s

Why Suitable:
It prevents the creation of multiple database connections and improves system performance and resource management.

4.3.2 Factory Pattern

The Factory Pattern is used to create different types of civic issue objects (e.g., Garbage Issue, Water Leakage Issue, Road Damage Issue) based on user input without exposing the object creation logic.

Why Suitable:
It improves flexibility, simplifies object creation, and makes the system easier to extend when new issue categories are added.

4.3.3 Observer Pattern

The Observer Pattern is used for the notification system. Whenever the status of an issue changes (Submitted, In Progress, Resolved), subscribed users automatically receive updates and notifications.

Why Suitable:
It enables real-time communication between the system and users while maintaining loose coupling between notification services and issue management modules.

5. Testing and Results

5.1 Test Cases

Test Case No.	Test Scenario	Expected Result	Observation
TC-01	User Registration	User account created successfully	Pass
TC-02	User Login	User logs in with valid credentials	Pass
TC-03	Report Civic Issue	Issue submitted successfully with location and image	Pass
TC-04	Track Issue Status	Current issue status is displayed correctly	Pass
TC-05	AI Issue Classification	System classifies issue into correct category	Pass
TC-06	Admin Assigns Issue	Issue assigned to appropriate department	Pass
TC-07	Department Updates Status	Status changes to In Progress/Resolved	Pass
TC-08	Notification Delivery	User receives status update notification	Pass

Test Case No.	Test Scenario	Expected Result	Observation
TC-09	**Feedback Submission**	**User feedback recorded successfully**	**Pass**
TC-10	**Generate Reports**	**Admin dashboard displays reports and analytics**	**Pass**

5.2 Results

The implementation of NagarMitram successfully achieved the objectives of providing a smart, efficient, and user-friendly civic issue management platform. The system enables citizens to report civic issues using images, descriptions, and GPS-based location information, while municipal authorities can monitor and manage complaints through a centralized dashboard.

Testing results showed that user registration, login, issue reporting, issue tracking, AI-based classification, department assignment, status updates, notifications, and feedback submission functions operated correctly. The AI module was able to categorize reported issues and assist in prioritizing complaints for faster resolution.

The administrative dashboard provided real-time visibility of complaints, allowing authorities to efficiently assign tasks and monitor progress. Notifications and status tracking improved transparency and citizen engagement throughout the complaint lifecycle.

Overall, the system demonstrated improved complaint management efficiency, reduced response time, enhanced accountability, and better communication between citizens and municipal departments. The generated UML diagrams, project screenshots, and test case results further validate the successful implementation of the proposed solution.

Performance Observations:

- Accurate issue reporting with GPS location tracking.
- Successful AI-based issue classification and prioritization.
- Real-time status updates and notification delivery.
- Efficient complaint assignment and monitoring through the admin dashboard.
- Improved transparency and citizen participation in civic governance.

6. Conclusion

The NagarMitram project successfully developed an AI-powered civic issue reporting and management platform that improves communication between citizens and municipal authorities. The system provides a convenient and transparent mechanism for reporting civic

problems such as waste management issues, water leakage, damaged roads, and streetlight failures through a mobile application integrated with GPS and image-based reporting.

The implementation incorporates modern technologies, real-time data processing, and Artificial Intelligence techniques for issue classification, prioritization, and efficient complaint management. The administrative dashboard enables authorities to monitor, assign, and resolve issues effectively while maintaining transparency throughout the process.

The successful testing and deployment of the system demonstrate its ability to streamline complaint handling, improve response times, enhance citizen participation, and support data-driven decision-making. The project contributes to the vision of smart governance by providing a scalable, user-friendly, and intelligent platform for civic issue management.

Overall, NagarMitram offers significant benefits including increased operational efficiency, improved accountability, better resource utilization, enhanced citizen satisfaction, and stronger collaboration between government agencies and the public, making it a valuable solution for modern urban governance.

7. Future Scope

7.1 Enhancement 1

IoT-Based Automatic Issue Detection

Future versions of NagarMitram can integrate IoT sensors and smart devices to automatically detect civic issues such as overflowing garbage bins, water leakages, air pollution, and streetlight failures, reducing dependence on manual reporting.

7.2 Enhancement 2

Multilingual and Voice-Based Support

The system can be enhanced with multilingual interfaces and voice-based complaint registration to improve accessibility for users from diverse linguistic backgrounds and varying levels of digital literacy.

7.2 Enhancement 3

Advanced AI and Predictive Analytics

Future development can incorporate advanced machine learning models to predict recurring civic issues, optimize resource allocation, analyze citizen sentiment, and recommend preventive actions for municipal authorities.

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